

Boss Challenge — Paper 03 (Class 7)

Motion & Time | Respiration in Organisms | Simple Equations

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Speed of a moving body is found by:
(A) distance / time
(B) distance + time
(C) distance x time
(D) time / distance
2. Breathing in (inhalation) brings into the body mostly:
(A) water vapour
(B) oxygen
(C) nitrogen for energy
(D) carbon dioxide
3. Solve: $x + 5 = 12$
(A) $x = 17$
(B) $x = 7$
(C) $x = 5$
(D) $x = 60$
4. The SI unit of time is the:
(A) kilometre
(B) metre
(C) second
(D) hour
5. Compared with the air we breathe in, the air we breathe out has more:
(A) dust
(B) oxygen
(C) carbon dioxide
(D) nitrogen
6. Solve: $3x = 21$
(A) $x = 18$
(B) $x = 7$
(C) $x = 24$
(D) $x = 63$
7. A car travels 60 km in 2 hours. Its speed is:
(A) 2 km/h
(B) 30 km/h
(C) 120 km/h
(D) 62 km/h

8. Respiration is important because it:

- (A) cools the blood
- (B) releases energy from food
- (C) removes bones
- (D) makes food

9. Solve: $x / 4 = 5$

- (A) $x = 1$
- (B) $x = 1.25$
- (C) $x = 9$
- (D) $x = 20$

10. Motion in which a body covers EQUAL distances in equal intervals of time is called:

- (A) non-uniform motion
- (B) circular motion
- (C) rest
- (D) uniform motion

11. During heavy exercise our muscles may respire without enough oxygen, producing:

- (A) alcohol
- (B) oxygen
- (C) glucose
- (D) lactic acid

12. Solve: $2x + 1 = 9$

- (A) $x = 5$
- (B) $x = 3$
- (C) $x = 16$
- (D) $x = 4$

13. The instrument that measures the total distance a vehicle has travelled is the:

- (A) clock
- (B) odometer
- (C) thermometer
- (D) speedometer

14. Fish breathe using their:

- (A) lungs
- (B) gills
- (C) skin only
- (D) fins

15. 'A number increased by 7 is 20.' The number is:

- (A) 13
- (B) 140
- (C) 3
- (D) 27

16. One complete to-and-fro swing of a pendulum is called one:

- (A) revolution
- (B) oscillation
- (C) rotation
- (D) rest

17. Insects (like a cockroach) breathe through tiny holes called:

- (A) gills
- (B) lungs
- (C) spiracles
- (D) stomata

18. Solve: $5x - 2 = 13$

- (A) $x = 2.2$
- (B) $x = 5$
- (C) $x = 11$
- (D) $x = 3$

19. The time taken by a pendulum to complete one oscillation is called its:

- (A) speed
- (B) frequency of light
- (C) distance
- (D) time period

20. Yeast respire without oxygen (anaerobically) to produce alcohol and:

- (A) water only
- (B) oxygen
- (C) lactic acid
- (D) carbon dioxide

21. Solve: $4x = -16$

- (A) $x = 4$
- (B) $x = -4$
- (C) $x = -64$
- (D) $x = -12$

22. On a distance-time graph, a straight slanting (sloping) line means the object is moving with:

- (A) zero speed
- (B) changing speed
- (C) uniform speed
- (D) backward motion

23. The dome-shaped muscle that helps us breathe is the:

- (A) stomach
- (B) diaphragm
- (C) liver
- (D) heart

24. Solve: $7 - x = 3$

- (A) $x = -4$
- (B) $x = 21$
- (C) $x = 4$
- (D) $x = 10$

25. If a pendulum makes 10 oscillations in 20 seconds, its time period is:

- (A) 2 s
- (B) 0.5 s
- (C) 10 s
- (D) 200 s

26. Earthworms breathe through their:

- (A) lungs
- (B) spiracles
- (C) gills
- (D) moist skin

27. Solve: $3x + 2 = 2x + 9$

- (A) $x = 11$
- (B) $x = 1$
- (C) $x = 5$
- (D) $x = 7$

28. Average speed of a journey is:

- (A) total distance / total time
- (B) distance of one part only
- (C) highest speed reached
- (D) total time / total distance

29. Plants take in oxygen for respiration mainly through tiny pores called:

- (A) roots only
- (B) stomata
- (C) spiracles
- (D) gills

30. Solve: $x/2 + 1 = 4$

- (A) $x = 3$
- (B) $x = 8$
- (C) $x = 10$
- (D) $x = 6$

31. The motion of the hands of a clock is an example of:

- (A) no motion
- (B) straight-line motion
- (C) random motion
- (D) periodic motion

32. Breathing and respiration are different because breathing is a physical process, while respiration:

- (A) makes food
- (B) only happens in the nose
- (C) is the same as breathing
- (D) releases energy inside cells

33. Solve: $x - 5 = -2$

- (A) $x = -3$
- (B) $x = 7$
- (C) $x = 3$
- (D) $x = -7$

34. A car covers 150 km in 3 hours. Its average speed is:

- (A) 3 km/h
- (B) 450 km/h
- (C) 50 km/h
- (D) 153 km/h

35. Exhaled air turns lime water milky. This shows the exhaled air contains:

- (A) nitrogen
- (B) carbon dioxide
- (C) oxygen
- (D) water

36. Solve: $6x = 6$

- (A) $x = 36$
- (B) $x = 0$
- (C) $x = 1$
- (D) $x = 12$

37. Distance is usually measured in:

- (A) kilograms
- (B) metres or kilometres
- (C) degrees Celsius
- (D) seconds

38. Which gas is essential for aerobic respiration?

- (A) hydrogen
- (B) nitrogen
- (C) oxygen
- (D) carbon dioxide

39. Solve: $x/3 = -2$

- (A) $x = 6$
- (B) $x = -2/3$
- (C) $x = 1$
- (D) $x = -6$

40. If the speed stays the same and the time doubles, the distance covered:

- (A) doubles
- (B) becomes zero
- (C) stays the same
- (D) halves

41. When we breathe in, the diaphragm:

- (A) turns into bone
- (B) moves down and the chest expands
- (C) moves up and chest shrinks
- (D) stops moving

42. Solve: $2(x + 1) = 10$

- (A) $x = 5$
- (B) $x = 4$
- (C) $x = 9$
- (D) $x = 3$

43. Periodic motion means motion that:

- (A) repeats at regular intervals of time
- (B) is always in a straight line
- (C) happens only once
- (D) never repeats

44. Anaerobic respiration releases ____ energy than aerobic respiration.

- (A) the same
- (B) more
- (C) less
- (D) no

45. 'Twice a number minus 3 is 7.' The number is:

- (A) 5
- (B) 8
- (C) 10
- (D) 2

46. A train moving at 72 km/h is ____ a car moving at 50 km/h.

- (A) slower than
- (B) the same speed as
- (C) at rest compared to
- (D) faster than

47. The tube that carries air from the throat down to the lungs is the:

- (A) backbone
- (B) food pipe
- (C) blood vessel
- (D) windpipe (trachea)

48. Solve: $8 - 2x = 2$

- (A) $x = -3$
- (B) $x = 5$
- (C) $x = 3$
- (D) $x = 6$

49. The faster a vehicle moves, the ____ its line on a distance-time graph.

- (A) flatter
- (B) steeper
- (C) more horizontal
- (D) more wavy

50. Aerobic respiration of glucose gives carbon dioxide, energy and:

- (A) lactic acid
- (B) oxygen
- (C) water
- (D) alcohol

51. Solve: $10 = 2x$

- (A) $x = 5$
- (B) $x = 8$
- (C) $x = 12$
- (D) $x = 20$

52. To cover 100 km at a steady speed of 50 km/h, the time needed is:

- (A) 0.5 hour
- (B) 200 hours
- (C) 2 hours
- (D) 50 hours

53. Gills are well supplied with blood so that they can:

- (A) filter food
- (B) help swimming
- (C) absorb dissolved oxygen from water
- (D) make scales

54. Solve: $x + 9 = 4$

- (A) $x = -13$
- (B) $x = 5$
- (C) $x = 13$
- (D) $x = -5$

55. A speedometer reading of 0 km/h means the vehicle is:
- (A) moving very fast
 - (B) out of fuel
 - (C) at rest (not moving)
 - (D) reversing
56. Why do we breathe faster after running?
- (A) to make food
 - (B) muscles need more oxygen to release energy
 - (C) to cool the lungs
 - (D) to add nitrogen
57. Solve: $5x = 45$
- (A) $x = 9$
 - (B) $x = 40$
 - (C) $x = 225$
 - (D) $x = 50$
58. If a cyclist covers 20 km in the first hour and only 10 km in the next hour, the motion is:
- (A) rest
 - (B) non-uniform
 - (C) circular
 - (D) uniform
59. Plants release more carbon dioxide than oxygen:
- (A) never
 - (B) only in winter
 - (C) at night
 - (D) at noon
60. Solve: $x/5 = 3$
- (A) $x = 0.6$
 - (B) $x = 2$
 - (C) $x = 15$
 - (D) $x = 8$
61. Which of these shows uniform motion?
- (A) a car covering 40 km every hour
 - (B) a car speeding up at a signal
 - (C) a stone thrown up and falling
 - (D) a swinging child
62. Cellular respiration takes place in:
- (A) only in animals
 - (B) only the heart
 - (C) all living cells
 - (D) only the lungs

63. Solve: $4x + 4 = 20$

- (A) $x = 4$
- (B) $x = 5$
- (C) $x = 24$
- (D) $x = 6$

64. Galileo first studied time-keeping using a swinging:

- (A) candle
- (B) wheel
- (C) spring only
- (D) pendulum

65. Which is the correct word equation for aerobic respiration?

- (A) glucose + oxygen $\rightarrow$ carbon dioxide + water + energy
- (B) oxygen $\rightarrow$ energy
- (C) carbon dioxide + water $\rightarrow$ glucose + oxygen
- (D) glucose $\rightarrow$ alcohol + CO $\blacksquare$

66. Solve: $3 = x - 6$

- (A) $x = 2$
- (B) $x = -3$
- (C) $x = 9$
- (D) $x = 18$

67. The needle of a speedometer rising steadily means the vehicle is:

- (A) reversing
- (B) at rest
- (C) slowing down
- (D) speeding up (accelerating)

68. An earthworm's skin must stay moist because:

- (A) to swim
- (B) it likes water
- (C) to stay warm
- (D) gases can only pass through a moist surface

69. Solve: $9 + x = 9$

- (A) $x = 0$
- (B) $x = 18$
- (C) $x = 9$
- (D) $x = 1$

70. Speed is measured in units like:

- (A) kilograms
- (B) metres per second (m/s) or km/h
- (C) metres only
- (D) seconds only

71. Breathing rate (breaths per minute) generally ____ during exercise.

- (A) increases
- (B) becomes zero
- (C) decreases
- (D) stays exactly the same

72. Solve: $2x - 7 = 5$

- (A) $x = -1$
- (B) $x = 1$
- (C) $x = 12$
- (D) $x = 6$

73. An object whose distance-time graph is a straight horizontal line is:

- (A) at rest
- (B) moving fast
- (C) accelerating
- (D) moving slowly

74. Which of these animals breathes through its skin?

- (A) earthworm
- (B) human
- (C) cockroach
- (D) fish

75. If $x = 4$, is it a solution of $2x + 3 = 11$?

- (A) No, it gives 14
- (B) Cannot be checked
- (C) No, it gives 9
- (D) Yes, because $2 \times 4 + 3 = 11$

76. The motion of a child on a swing is best described as:

- (A) oscillatory (to-and-fro)
- (B) straight-line motion
- (C) circular motion
- (D) no motion

77. Anaerobic means respiration that happens:

- (A) only in the lungs
- (B) with lots of oxygen
- (C) only in plants
- (D) without oxygen

78. Solve: $x - 7 = 0$

- (A) $x = -7$
- (B) $x = 7$
- (C) $x = 0$
- (D) $x = 14$

79. Two runners start together; the one with the higher speed will:

- (A) never finish
- (B) take more time
- (C) reach the finish first
- (D) reach last

80. The exchange of gases in the lungs happens in tiny air sacs called:

- (A) spiracles
- (B) villi
- (C) alveoli
- (D) stomata

81. Solve: $x/4 = -3$

- (A) $x = -7$
- (B) $x = 12$
- (C) $x = -0.75$
- (D) $x = -12$

82. Which quantities must you know to calculate speed?

- (A) mass and time
- (B) distance and time
- (C) time only
- (D) distance only

83. Which process is almost the reverse of photosynthesis?

- (A) transpiration
- (B) aerobic respiration
- (C) digestion
- (D) breathing

84. Solve: $12 = 3x$

- (A) $x = 15$
- (B) $x = 36$
- (C) $x = 4$
- (D) $x = 9$

85. An odometer of a car reads 25,430 km at the start and 25,480 km at the end of a trip. Distance travelled:

- (A) 50 km
- (B) 50,910 km
- (C) 0 km
- (D) 25,455 km

86. Breathing is to moving air as respiration is to:

- (A) digesting food
- (B) pumping blood
- (C) releasing energy from food
- (D) making food

87. Solve: $2x + 2x = 16$

- (A) $x = 16$
- (B) $x = 4$
- (C) $x = 8$
- (D) $x = 2$

88. The to-and-fro time of a pendulum stays the same as long as we don't change its:

- (A) how loud the room is
- (B) colour
- (C) length
- (D) the day

89. Which part lets a fish take in oxygen from water?

- (A) gills
- (B) fins
- (C) scales
- (D) tail

90. Solve: $7x = 0$

- (A) $x = 0$
- (B) $x = 7$
- (C) $x = 1$
- (D) $x = -7$

91. A bus covers 80 km in the first 2 hours and 80 km in the next 2 hours. Its motion is:

- (A) circular
- (B) uniform
- (C) at rest
- (D) non-uniform

92. Which animal breathes using lungs?

- (A) a cockroach
- (B) an earthworm
- (C) a fish
- (D) a dog

93. Solve: $x + 0 = -4$

- (A) $x = 4$
- (B) $x = -8$
- (C) $x = 0$
- (D) $x = -4$

94. The time period of a pendulum is measured in:

- (A) seconds
- (B) km/h
- (C) kilograms
- (D) metres

95. Bread dough rises because yeast respire and releases:

- (A) carbon dioxide gas
- (B) nitrogen
- (C) oxygen
- (D) hydrogen

96. Solve: $3x - 1 = 8$

- (A) $x = 27$
- (B) $x = 3$
- (C) $x = 2$
- (D) $x = 9$

97. If distance is in kilometres and time in hours, the speed unit is:

- (A) km/h
- (B) km
- (C) kg
- (D) hours

98. We should breathe through the nose because it:

- (A) filters, warms and moistens the air
- (B) makes food
- (C) pumps blood
- (D) cools the brain directly

99. Solve: $2x = -10$

- (A) $x = -5$
- (B) $x = -8$
- (C) $x = -20$
- (D) $x = 5$

100. A car going around a roundabout while its speed keeps changing shows:

- (A) uniform motion
- (B) straight-line motion
- (C) non-uniform motion
- (D) rest